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Prioritizing-based Message Scheduling for Reliable Unmanned Aerial Vehicles Ad Hoc Network

Jun Li[a,c,*], Ming Chen[a,b], Fei Dai[a], and Huibing Wang[a,c]

[a]*College of Command Information Systems, Army Engineering University of PLA, Nanjing, 210000, China*
[b]*College of Computer Science and Technology, Nanjing University of Aeronautics and Astronautics, Nanjing, 210000, China*
[c]*School of Computer and Information, Anhui Polytechnic University, Wuhu, 241000, China*

Abstract

The Unmanned Aerial Vehicles Ad hoc Network (UAANET) consists of multi UAVs through multi-hop wireless communication links, and it can execute missions more efficiently compared with the single UAV. Due to its dynamic topology, fast-moving velocity, and unstable radio channel quality, message delivery in UAANET often suffers from increased delays and packet loss. In particular, the command messages and the coordination messages are lost or significantly delayed, which results in uncontrolled UAV, failure of the task execution, or even the crash of UAANET. For the reliability and performance of UAANET, the message scheduling scheme should be taken into account. In this paper, we propose a Prioritizing-based Message Scheduling algorithm (PMS) to provide reliable transmission of command messages and coordination messages. The proposed algorithm assigns different priorities to the messages based on the content of messages and dynamic context factors, and then schedules the messages respectively. Simulation results verify that PMS can substantially increase the reliability of UAANET.

Keywords: UAANET; message scheduling priority; UAV; reliability

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1. Introduction

Throughout the last decade, the UAV has been utilized extensively for civilian applications due to its low cost, rapid deployment, and flexibility. The UAV is involved in many civilian areas like border patrolling, environment surveillance, agriculture and forest monitoring, package delivery, search and rescue, infrastructure inspection, and disaster communication network [1-2]. Because of its limit payload volume and flight time, a small UAV has constraints on its computing ability, numbers and types of sensors, battery capacity, which can be an issue for the complex mission. Individual UAV itself also is a single point of failure in the duration of the task execution, which is unacceptable for the most civilian applications of UAV [1, 3].

An alternative method is to use multiple UAVs, which are more efficient and reliable, and also carry out missions that are beyond the capabilities of the individual UAV, owing to more numbers and types of sensors, extensive coverage, fault tolerance, and concurrent execution of the task. To allow for collaboration and coordination between UAVs, multiple UAVs must have the ability to communicate with each other through a multihop wireless ad hoc network called the unmanned aerial vehicle ad hoc network (UAANET) [1, 4]. As illustrated in [1, 4-5], UAANET is a particular subclass of Mobile Ad hoc Network (MANET) characterized by high node moving velocity, faster-changing network topology, and 3D mobility. Except for the above-described features, messages transferred in UAANET not only include application messages such as videos, images, temperature, humidity, and audio, but also involve command messages and coordinating messages [6], unlike the other Mobile Ad hoc Networks. Command messages consisting of mission commands, mission statuses, and telemetry data are critical for the effective control of multiple UAVs. Coordination messages are proven useful for coordinating the efficient collaborative task execution of multiple UAVs and collision avoidance [6]. Therefore, the

* Corresponding author .
E-mail address: edmondlee@ahpu.edu.cn

successful delivery of the command messages and coordination messages is a vital issue for guaranteeing UANNET reliability.

In the practical context, UAANET faces various challenges that impact its performance and reliability. One challenge is congestion. If there is a significant number of packets transferred through UAANET, then UAANET experiences congestion, which leads to increased delays, jitter, and packet loss. Another challenge is the dynamic nature of UAANET, owing to its quick-change topology, node mobility, and obstacle avoidance. The distance between neighbor nodes of UAANET is frequently changed. The wireless link bandwidth is relative to the distance between the adjacent nodes, so the link bandwidth of UAANET is highly dynamic. Due to the above issues, successful transmission of all messages cannot be ensured without an efficient message scheduling scheme. Command messages and coordination messages are especially lost or severely delayed, which results in uncontrolled UAVs, failure of the task execution, or even the crash of UAANET. Thus, reliable transmission and QoS guarantee of the command messages and the coordination messages are critical for the reliability of UAANET.

There have been various papers focused on multiple UAVs or UAANET foci on routing algorithm, task assignment, autonomy, and cooperation [1, 3-5, 7]. Few papers are about congestion control and prioritizing-based packet scheduling in UAANET. In this paper, the novel message scheduling algorithm is presented to guarantee the delivery of the command messages and coordination messages. The message scheduling algorithm first assigns priority to messages based on the content of the message, message size, and dynamic wireless link quality. Then, the proposed algorithm schedules the prioritized message in the queue. The performance of our proposed algorithm is evaluated using various performance metrics including average end-to-end delay, packet loss ratio, and average throughput.

The rest of the paper is structured as follows. In Section 2, papers about UAANET are reviewed favorably. In Section 3, we present the prioritize-based message scheduling scheme and its features in detail. In Section 4, the proposed scheduling algorithm is implemented in the NS-3 simulation environment, and performance evaluations regarding the average end-to-end delay, average throughput, average queuing delay, and packet loss ratio are presented. Finally, we discuss the obtained results in Section 5.

2. Related Works

In recent years, more attention has concentrated on the provision of routing algorithms for UAANET to support efficient data delivery. Because UAANET is considered as a particular case of MANET and VANET, most present papers adapt traditional MANET routing algorithms to satisfy the requirements of UAANET [5, 7].

P-OLSR [8] introduced the Speed-Weighted ETX (Expected Transmission Count) metric to predict the wireless link quality, which combines the wireless link condition, speed, direction, and position of the UAV. P-OLSR uses this metric to choose the neighbor node to cope well with the fluid topology of UAANETs. In literature [9], Y. Zheng et al. proposed the ML-OLSR routing algorithm that uses the mobility aware algorithm to select the most stable node as the next relay and utilizes the load-aware algorithm to decrease transmission delay and increase the packet delivery ratio under collision condition. P-OLSR and ML-OLSR are both extensions of the OLSR routing protocol. OLSR implies that each node exchanges topology control and hello packets with neighbor nodes, introducing control overhead. In UAANET, with frequent network topology changes, the route lifetime rapidly decreases and the P-OLSR and ML-OLSR recovery route is frequent. This results in significant control overhead.

RGR [10] assumed a UAANET routing algorithm that combines the reactive routing algorithm and geographical routing algorithm. RGR uses AODV first to forward the packet; if there are route breaks, RGR switches into the GGF forwarding algorithm to relay packets to the nearest neighbor and the destination. The RGR routing algorithm achieves good performance in UAANETs, as it can switch between AODV and GGF according to dynamic network topology. However, the drawback of RGR is that increased control overhead leads to network congestion. In [11], the RARP routing scheme was proposed for UAANETs. The RARP routing scheme combines unicasting and geocasting routing based on location and trajectory information. It improves the performance of UAANETs by using dynamic angle adjustment transmission and keeping track of changing network topology. The weakness of RAPP is its introduction of heavy computing and storage for UAV.

The above-discussed routing algorithms can be classified into two categories: topology-based routing and geography-based routing. Topology-based routing and geography-based routing both relay all packets equally and do not consider respective QoS requirements of diverse types of messages in UAANETs. These routing algorithms cannot offer reliable transmission of the command messages and the coordination messages under network congestion and dynamic link quality.

Unlike the above-discussed routing algorithm, Qi et al. [12] offered a centralized Traffic-Differentiated Routing (TDR) algorithm with SDN Cluster controller and a coordination controller. The routing algorithm groups multiple UAVs into several cluster domains; each cluster domain includes a cluster SDN controller, which schedules network resources and guides data forwarding. A stationary airship coordination controller realizes the interaction between the single-domain controllers. A noticeable advantage of the routing algorithm is that it introduces SDN into UAANET to provide scalability, flexibility, and efficiency. Another advantage of the routing algorithm is that it offers TDR algorithm in each cluster to ensure the different QoS requirements of delay-sensitive applications and reliability-requisite applications in UAANETs. The routing algorithm focuses on how to meet the QoS requirements of diverse applications in UAANET, which cannot provide reliable transmission and QoS guarantee of the command messages and coordination messages delivered in UAANETs, facing network congestion or degradation of wireless link quality.

In literature [13], Ouns Bouachir et al. proposed an ad hoc network QoS architecture for cooperative UAVs, called DAN. The DAN scheme uses queueing and scheduling strategies to differentiate traffic and is classified into three classes: Urgently, Premium, and Best-Effort. The defect of DAN is that it does not consider the moving speed of UAV, dynamic topology, or quality of radio channel, which impairs the successful delivery of messages in UAANET.

3. The Design of PMS Algorithm

Because UAANET faces unstable link quality, frequently changing topology, and 3D mobility, guaranteeing the delay constraint and QoS quality of command messages and coordination messages is a critical issue for UAANET. Due to the dynamic application environment, assigning priority to messages and scheduling prioritized messages is a challenge in UAANETs. In this section, we propose a prioritizing-based message scheduling algorithm to efficiently prioritize and schedule messages of command and coordination, enhancing the performance and reliability of UAANETs. To assign the priority to messages in the queue, we should consider various factors like the content of messages, message length, quality of radio channels, and so on.

Figure 1 shows the schematic of the PMS algorithm. The PMS includes two modules: Message Priority Assignment module and Message Scheduling module. The Message Priority Assignment module defines the priority of each message in the queue, based on the size and content of the message, and the dynamic context factor. The Message Scheduling module reschedules the prioritized messages in the output queue. Every node in UAANET independently prioritizes and schedules each message to improve the reliability and performance of the entire network.

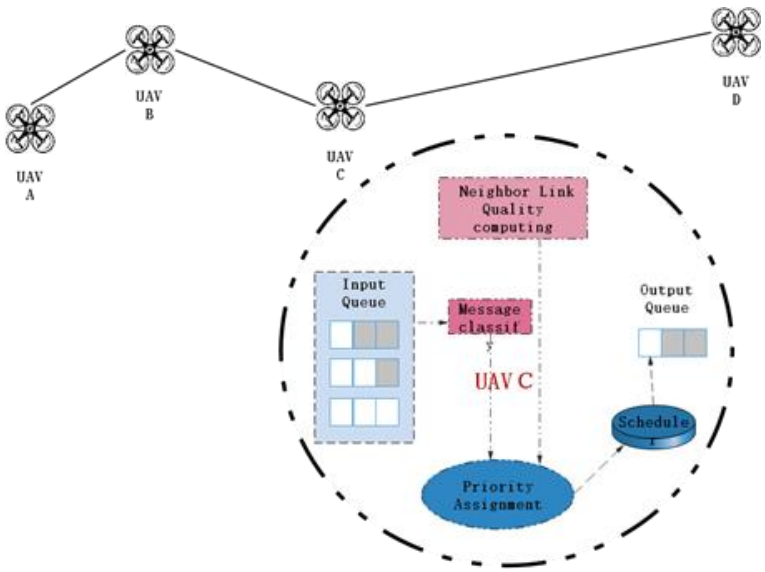

Figure 1. Schematic of the prioritizing-based message scheduling algorithm

3.1. Message Priority Assignment Module

In the Message Priority Assignment module, messages are assigned by different priorities based on the content of messages and environment context. Then, the messages in the output queue are scheduled based on assigned priorities to meet the

QoS requirement of different class messages. We assume that the priority of each message is defined based on message static factors and environment context factors as well as message size.

$$Priority_{message} = \frac{Factor_{message} \times Factor_{context}}{size\ of\ \ message} \tag{1}$$

$Priority_{message}$ is directly related to $Factor_{message}$ and $Factor_{context}$. Because the size of the command message and coordination message is smaller than the application and sensing messages, $Priority_{message}$ is opposite to the size of the message. The $Factor_{message}$ for a message is considered to three priority categories, including $priority_{application}$, $priority_{coordination}$, *and* $priority_{command}$, which are based on the content of messages. Each priority category is defined as follows:

- $Priority_{application}$ is designed for the application messages. The application messages include two categories: the first category of application messages is generated by onboard sensors of the UAV node, like temperature data, humidity data, and infrared data. These messages are delay tolerant and periodic. They usually suffer long delays, without impairing the application of the UAANET. Application messages in the second category have high throughput and do not tolerant significant delays and jitter, such as video, audio, and live video streaming. These messages fail in transmission or frequently retransmit, which impairs the QoE of the end user.
- $Priority_{coordination}$ is designed for coordination messages and is focused on the interaction between UAV nodes. The coordination messages are used for task allocation, task monitoring, collaboration, and collision avoidance among UAVs. The coordination information is key to successful mission execution.
- $Priority_{command}$ is considered for the UAV's command message and consists of waypoint control, mission control, remote control, mission status, and telemetry data. The transmission delay of command messages must be guaranteed under an acceptable threshold value. The failure of command message transmission will degrade the reliability of UAANET and may even make the entire UAANET crash. The crash of UAANET could lead to the impairment of people's safety and the loss of ground property.

The $Factor_{context}$ is defined to illustrate how the variety of context factors influence the performance and reliability of UAANET. We propose three metrics to evaluate the impact of the context factors, including the expiry of the message metric (EM), the overlapping of the UAV node's waypoint metric (OUW), and the neighbor link quality metric (NLQ). These metrics are defined as follows:

1) EM metric: The EM metric indicates the validity of the message. The difference between $t_{deadline}$ and t_{now} is to define the remaining time of the message. The value of t_{trans} shows an estimated time to send the message between the sender node and the receiver node. As defined by Equation (2), the message should be assigned the high priority when the value of EM is small. Therefore, the priority of the message is opposite to the EM metric.

$$EM = \frac{t_{deadline} - t_{now}}{t_{trans}} \tag{2}$$

2) OUW metric: The overlapping of the UAV node's waypoint is considered as a metric for evaluating the successful message transmission probability between the sender UAV node and receiver UAV node. At waypoint w_k the sender node n_i and the receiver node n_j respectively have three-dimensional positions $\{x_i, y_i, z_i\}$ and $\{x_j, y_j, z_j\}$, and the distance $d_{i,j}(x)$ between n_i and n_j can be computed by Equation (3).

$$d_{i,j}(w_k) = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2 + (z_j - z_i)^2} \tag{3}$$

$$I(w_k) = \begin{cases} 1, \text{ while } d_{i,j}(w_k) \le R_{max_comm} \\ 0, \text{ while } d_{i,j}(w_k) > R_{max_comm} \end{cases} \tag{4}$$

$$OUW_{(i,j)} = \frac{\sum_{w_k} I(w_k)}{N} \quad , \ w_k \in W, \ W = \{w_1, w_2, w_3, \cdots, w_N\} \tag{5}$$

According to Equation (4), if distance $d_{i,j}(x)$ is less than the maximum communication radius R_{max_comm}, the value of $I(w_k)$ is 1; otherwise, the value of $I(w_k)$ is 0. Thus, at a series of waypoint w_1, w_2, w_3, ⋯, w_N in the future, the $OUW_{(i,j)}$ of node n_i and node n_j is computed by Equation (5). When the metric OUW is high, the sender node and receiver node have a higher chance of transmitting or retransmitting messages. Therefore, the probability of successful transmission between the sender and receiver node is high. Even if the message once failed in transmission, it has an increased chance to be retransmitted and successfully received by the receiver node, so a low priority should be assigned to messages with a high OUW metric.

3) NLQ metric: We use the neighbor link quality metric to represent the wireless link condition between the UAV node and its one-hop neighbor nodes. The NLQ metric is similar to ETX, using the forward receiving ratio and the reverse receiving ratio to evaluate the link quality. Compared to the ETX metric, which computes the ETX value of the whole route from the source node to the destination node, the NLQ metric is proposed to estimate the link quality among the sender node and its one-hop neighbor nodes. The low value of NLQ indicates that the sender UAV node has good wireless link quality among it and its neighbor nodes. Conversely, the high value of NLQ reveals that the wireless link quality is impaired by interference, congestion, and obstacles. With the high NLQ metric, the radio channel between the sender and receiver node will suffer higher packet loss and significant delays. The PMS suggests that the priority of the message is directly proportional to the value of NLQ. The value of NLQ metric is computed as follows:

$$NLQ = ETX(K_x) = \sum_{n_j \in K_x} \frac{1}{\phi(n_j)\omega(n_j)} \tag{6}$$

In Equation (6), K_x is a one-hop neighbor nodes set of the UAV node x and $\phi(n_j)$ is the forward receiving ratio, which indicates the probability that a packet sent to the node n_j is successfully received. $\omega(n_j)$ is the reverse receiving ratio, which presents the probability that the ACK packet is successfully received.

By combining Equations (2) and (6), the $Factor_{context}$ is calculated by Equation (7):

$$Factor_{context} = \frac{e^{NLQ}}{(EM+1)\times(OUW+1)} \tag{7}$$

In Equation (7), the EM metric and OUW metric are both added to 1 to avoid ambitious results when EM is equal to zero or OUW is equal to zero. Finally, based on (7) and (1), the priorities of the message are calculated by combing $Factor_{message}$, $Factor_{context}$, and the size of the message.

3.2. Message Scheduling Module

In the message scheduling module, the queue is rescheduled before transferring to the radio channel, based on the message priorities calculated in the Message Priority Assignment module. It is carried out as follows: Firstly, the algorithm calculates the load-based information d_{load} of the queue, and when the node d_{load} reaches half of the queue, it starts the message scheduling process. Equation (8) defines d_{load} as a ratio of two parameters: P_{in} and P_{out}.

$$d_{load} = \frac{P_{in}}{P_{out}} \tag{8}$$

The P_{in} metric shows the numbers of data packets that arrive at the node and the P_{out} metric indicates the numbers of data packets transferred from the node. Secondly, the packets in the queue are recalculated by the Message Priority Assignment module and reordered descending based on their priorities, when a new packet is sent to the queue. Finally, the packets are transferred from the queue to the wireless channel.

4. Simulation and Performance Evaluation

4.1. Simulation Setup

We implemented our algorithm in the NS3 network simulator, using 3D Obstacle Compliant Mobility Models [14]. The 3D Obstacle Compliant Mobility Models is proposed to simulate the environment where UAVs are deployed in reality, such as

urban areas with buildings or mountains. When the obstacles are introduced in our simulation, they result in the degradation of performance of UAANET, such as packet loss and increased end to end delay. We assume a prototype of UAANET that includes 40 UAV nodes and a mission area of 2km×2km. The moving speed of the UAV node is from 5 to 20m/s. We use the ML-OLSR algorithm as our prototype routing protocol to select the next relay node. The parameters of the simulation environment are listed in Table 1.

Table 1. Parameter setting of the simulation of UAANET

Parameter	Value
Number of UAV nodes	40
Simulation area	2km×2km
Node mobility model	obstacle compliant mobility [14]
UAV movement speed	5-20m/s
Routing protocol	ML-OLSR [9]
Channel capacity	10Mbps
Number of obstacles	0-20
Maximum transmission range	300m
Simulation time	200s

In the simulation process, we often add obstacles at random locations to trigger UAV avoidance events. When a single UAV avoids obstacles, the distance between it and its neighbors changes drastically, leading to dynamic fluctuations in network topology and radio link quality. In such a scenario, the proposed scheduling algorithm is assumed to guarantee the transmission of command messages and coordination messages. The performance of the PMS algorithm is compared with the FIFO scheduling strategy, regarding the average end-to-end delay, average throughput, average queuing delay, and packet loss ratio.

4.2. Simulation Result

The average end-to-end delay is the average time from the sender node to the receiver node. By tracking the average end-to-end delay, we evaluate the performance of PMS and FIFO during the periods of 0 to 20 obstacles, which are added in the mission area. Figure 2 shows the variations of the average end-to-end delay resulting from the different scheduling algorithms with the different number of obstacles in the mission area. With the increased number of obstacles, the average end-to-end delay increases for both PMS and FIFO. However, Figure 2 represents that the average end-to-end delay resulting from PMS is less than FIFO. The reason for this result is that PMS prioritizes and schedules messages and considers the content of the message, message size, and dynamic context factors. In contrast to the PMS algorithm, FIFO dequeues the packets without any prioritizing, which cannot guarantee the message delivery in dynamic topology and the dynamic link quality environment.

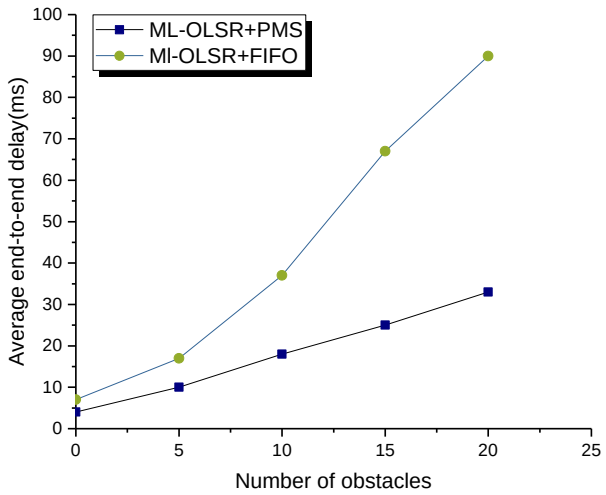

Figure 2. Impact of the number of obstacles on average end-to-end delay

The average throughput is defined as the amount of data transmitted over commutation channels per time slot. Figure 3 reveals the relationship between the average throughput and the number of obstacles in the mission area. The results indicate that the PMS algorithm outperforms the FIFO algorithm and improves the performance of UAANET. Initially,

there are no obstacles in the mission area. The average throughput of the PMS algorithm is equal to 8.6 Mbps, which is better than the FIFO algorithm, according to the dynamically scheduling of messages in the queue. As more and more obstacles are introduced in the mission area, the dynamic topology and worsened radio link quality impair the performance of the whole network. The PMS algorithm still mitigates the average throughput loss caused by obstacles compared with the FIFO algorithm.

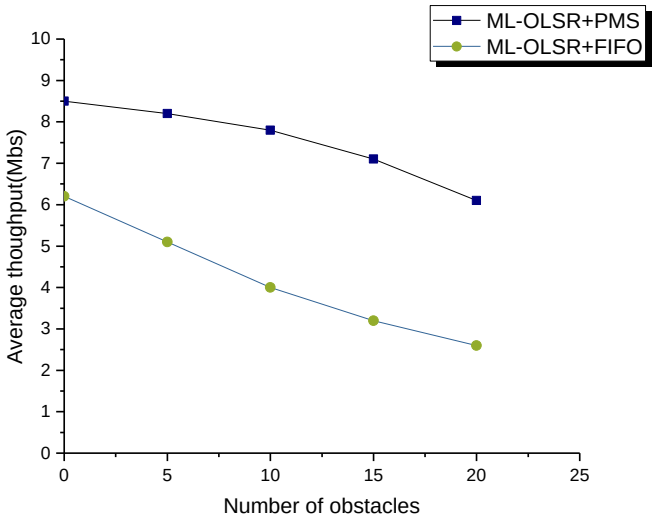

Figure 3. Variation of the average throughput with number of obstacles

The average queuing delay is the average time that the packet should wait in the output queue before being transferred to the wireless channels. In Figure 4, the impact of the number of obstacles on the average queuing delay is evaluated for all categories of messages. Figure 4 illustrates that the average queuing delay in the queue for command messages and coordination messages is much lower than the average queuing delay for application messages. The simulation result shows that the PMS algorithm transfers the command messages and coordination messages with low average queuing delay by prioritizing and rescheduling these messages in the queue, which provides reliability for UAANET.

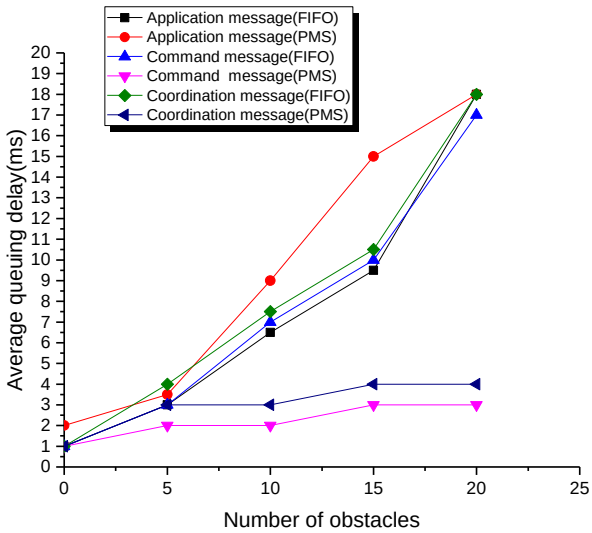

Figure 4. Impact of the number of obstacles on average queuing delay

The packet loss ratio is defined as the ratio of the number of packet loss to the number of successfully transmitted packets. Figure 5 shows the impact of the number of obstacles on the packet loss ratio of the command message, coordination message, and application message. The result reveals that packet loss ratio of the command message and coordination message is less than the packet loss ratio of the application message.

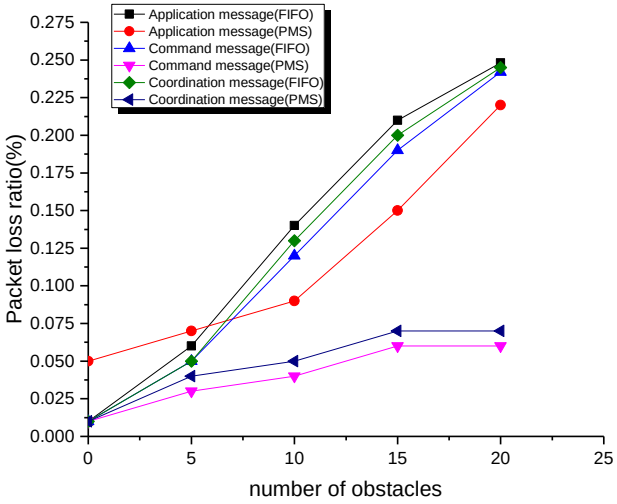

Figure 5. Impact of the number of obstacles on packet loss ratio

5. Conclusions

In this paper, to provide the reliable transmission of command messages and coordination messages in the UAANET, we proposed the PMS algorithm. Firstly, the proposed algorithm calculated the priority of each message based on the size of the message, the content of the message, and the dynamic context factor. Then, these messages were sent to the output queue and rescheduled for transferring to the wireless channel, based on the priority of messages. The performance of the PMS algorithm was evaluated and compared with the FIFO scheduling algorithm regarding average end-to-end delay, average throughput, average queuing delay, and packet loss ratio. The results indicated that application of the PMS algorithm improved the performance of UAANET by increasing the average throughput and reducing the average end-to-end delay as well as increased the reliability of UAANET by lowering the average queuing delay and packet loss ratio of the command message and coordination message. Therefore, the proposed PMS algorithm enhances the reliability of UAANET.

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Jun Li is currently pursuing a Ph.D. at Army Engineering University of PLA, Nanjing, China. He is also working as a lecturer at the School of Computer and Information, Anhui Polytechnic University, Wuhu, China. His research interests include UAV ad hoc network, software-defined wireless networking, and vehicular ad hoc network.

Ming Chen is currently a professor and Ph.D. supervisor at the College of Computer Science and Technology, Nanjing University of Aeronautics and Astronautics, Nanjing, China. He is also a Ph.D. supervisor at the Army Engineering University of PLA. His research interests include UAV ad hoc network, network performance analysis, and software-defined networking.

Fei Dai is currently a Ph.D. candidate at the Army Engineering University of PLA, Nanjing, China. His research interests include network measurement and monitoring, performance evaluation, and software-defined networking.

Huibing Wang is currently a Ph.D. candidate at the Army Engineering University of PLA, Nanjing, China. He is also a lecturer at the School of Computer and Information Engineering, Chuzhou University, Chuzhou, China His research interests include UAV ad hoc network, mobile ad hoc network, and software-defined networking.